A Parallel Sweep Ordering in SORh

This appendix expands the distinction between the block and wavefront SORh sweep modes used in Section 1.1. The distinction is mathematical as well as operational. For the linear discrete problem, the SORh update with lexicographic sweep ordering is an AOR iteration with

$$r_A = 2\omega_h, \qquad \omega_A = 2\omega_h \mathrm{CFL}_h. \tag{2}$$

At $\mathrm{CFL}_h = 1$, this reduces to classical SOR with relaxation parameter $2\omega_h$. As with Gauss–Seidel or SOR, the update order determines which components of the new iterate enter each row and therefore determines the iteration matrix.

For a southwest-to-northeast sweep, each point requires new values from its west and south predecessors and old values from its east and north neighbors. On a single mesh, this defines a partial order: all points on one southwest– northeast diagonal are independent once the preceding diagonal is complete. Figure 2 illustrates how the two modes treat that partial order when the mesh is divided into blocks.

A.1 Block mode

Block mode starts an ordered sweep in every locally ready block. Within a block, new west and south values are used immediately, so the local update has the intended SORh ordering. Across a block boundary, however, the neighboring block can be at the same stage of its own sweep. The incoming boundary value therefore belongs to the preceding global exchange rather than to the current ordered sweep. Equivalently, block mode replaces parts of the global Gauss–Seidel splitting with block-Jacobi couplings.

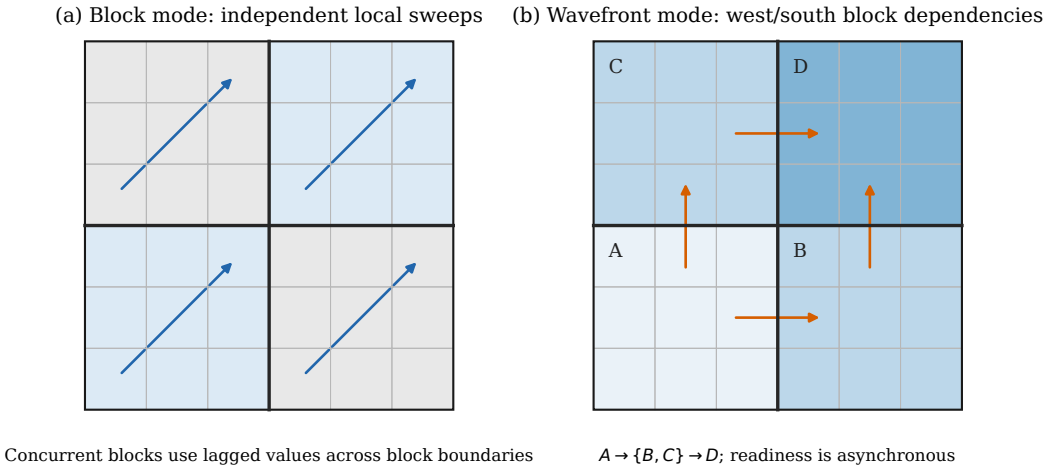


Figure 2: Conceptual comparison of SORh sweep modes. Block mode uses independent local sweeps and therefore lags cross-block data. Wavefront mode follows the dependency graph shown: A publishes to B and C, and D becomes ready after both have published. Ready blocks advance without a global diagonal barrier.

The benefit is concurrency: blocks can advance with few ordering constraints, and increasing the MPI rank count distributes these independent units well. The cost is that the effective iteration changes with the number and size of blocks. In the 128^2 stability study, the largest parameters used for the full Pancake trajectory were $\omega_h = 0.87$, 0.93, and 0.96 for 8^2-, 16^2-, and 32^2-cell blocks. The trend is expected: a larger block contains more of the global ordered sweep before encountering a lagged boundary and can therefore approach the global optimum. A short five-step screen suggested $\omega_h = 0.88$ for the smallest blocks, but that value diverged over the full trajectory; this is why the production benchmark used 0.87.

Block mode should consequently be viewed as a block-ordered variant of SORh, not simply as a communication-optimized realization of an invariant global iteration. Its converged solution remains the solution of the discrete Poisson equation if convergence is driven sufficiently far, but its finite-cap trajectory and stable parameter range depend on the decomposition.

A.2 Wavefront mode

Wavefront mode retains the global dependency graph. A block is released only after its current-sweep west and south faces arrive. Within each released block, cells follow the same i-fastest lexicographic order as in the serial sweep. The block then publishes its east and north faces, potentially releasing downstream blocks; blocks on the same diagonal front can execute concurrently. The periodic seam serves as the cut that starts the ordered sweep, but changing rank ownership or block layout does not change the order represented by the dependency graph.

Because this mode preserves the global iteration, the relaxation parameter is computed from the full mesh rather than from a block dimension. For the square 128^2 mesh,

$$\omega_h^* = \frac{1}{1 + \sin(\pi/128)} = 0.9760466. \tag{3}$$

This value was stable for every tested layout. At a high iteration cap, the wavefront profiles from 8^2-, 16^2-, and 32^2-cell blocks and from one and four MPI ranks agreed to relative L_2 differences below 1.7×10^{-13}. This is the practical evidence that the scheduling changes parallel execution without changing the represented iteration.

A.3 Measured consequences

Table 4 compares the two modes with 16^2-cell blocks, the same residual tolerance, a cap of 300 sweeps, and a 15-sweep residual-check interval. Each mode uses its stable relaxation parameter: 0.93 for block mode and 0.97605 for wavefront mode. Block mode has stronger rank scaling, but wavefront is substantially faster in absolute time because it requires fewer sweeps on average and uses the larger global parameter. It is also much closer to the converged MG profile at the finite cap.

Table 4: Block and wavefront SORh at $m_{\max} = 300$ with 16^2-cell blocks. Times are three-repetition medians. The profile column is the largest final relative L_2 difference from matched MG among five recorded fields.

Ranks	Mode	Wall (s)	Poisson (s)	Avg. sweeps	Converged	Profile diff.
1	Block	51.75	36.609	288.1	25/417	7.732%
1	Wavefront	27.43	12.580	279.5	48/417	0.224%
4	Block	22.96	17.270	288.1	25/417	7.732%
4	Wavefront	14.05	8.405	279.5	48/417	0.224%

The block-mode wall-time speedup from one to four ranks is 2.25 (56.3% parallel efficiency), compared with 1.95 (48.8%) for the matched wavefront case. Nevertheless, wavefront reduces absolute wall time by 47.0% with one rank and 38.8% with four ranks; its Poisson-time reductions are 65.6% and 51.3%. These results capture the central tradeoff. Block mode maximizes immediate parallel freedom, but pays for that freedom through a modified iteration and a smaller stable ω_h. Wavefront constrains the schedule enough to preserve the global SORh/AOR iteration while still exposing diagonal parallelism. For the Pancake benchmark, that balance produces the better absolute performance and the more stable finite-cap solution.

The evidence here is limited to a uniform two-dimensional mesh and one-to-four rank intranode execution. On adaptive meshes or much larger rank counts, front startup, load imbalance, and communication distance may change the balance. The ordering argument remains applicable, but those regimes require separate measurement.